

$$0111 = 7 = 2^3 - 1 = 7$$

Farmingdale State College
EET 223T Digital Electronics
Test #2

Date: Nov 5th, 09
Time: 1hr

Max Marks: 30

Name: _____

1. Represent the signed decimal using 2's complement system. Use 8 bits including a total of eight bits. (2 Marks)

a. +57 00111001
b. -74 10110110

$$\begin{array}{r} 01001010 \\ 10110101 \\ + 1 \\ \hline 10110110 \end{array}$$

2. The following numbers represents a signed decimal number in the 2's complement system. Determine the decimal number. (2 Marks)

a. 11011010 -38
b. 10000000 -128

3. What is the range of signed decimal value that can be represented using 16 bits, including the sign bit. Also determine the range of unsigned decimal values that can be represented using 16 bits. (2 Marks)

→ -2^{15} to $+2^{15}-1$ i.e. -32,768 to +32,767

Unsigned 0 → 65,535

4. How many bits would be needed to represent -4096 to +4095 (1 Mark)

12 bits for magnitude + 1 bit for sign → 13 bits

5. Perform addition by representing the numbers in 2's complement system (3 Marks)

a. Add +27 to -59
b. Add -37 to -67

5a)

$$\begin{array}{r} 00011011 + 27 \\ 01000101 - 59 \\ \hline 11100000 - 32 \end{array}$$

5b)

$$\begin{array}{r} 00100101 \rightarrow 11011011 \\ 01000011 \rightarrow 10111101 \\ \hline 11011011 - 37 \\ 10111101 - 67 \\ \hline 10011000 - 104 \end{array}$$

Disregard carry

$$\begin{array}{r} 01110111 \\ 10001001 \\ \hline \end{array}$$

6. Perform Subtraction by representing the numbers in 2's complement system (3 Mark)

a. Subtract +47 from +13
b. Subtract -37 from +95

6a)

$$\begin{array}{r} 00001101 \quad 13 \\ 11010001 \quad -47 \\ \hline 11011110 \quad -34 \end{array}$$

6b)

$$\begin{array}{r} 01011111 \quad 95 \\ 00100101 \quad -37 \\ \hline 10000100 \quad 132 \end{array}$$

↑ overflow

Not possible to represent in 8 bits

convexity

7. Add the following decimal numbers after them in to BCD code (1 Mark)

a. Add 987 to 123

$$\begin{array}{r} 1001 \ 1000 \ 0111 \\ 0001 \ 0010 \ 0011 \\ \hline 1010 \ 1010 \ 1010 \\ \hline 1110 \end{array} \quad \begin{array}{r} 987 \\ + 123 \\ \hline 1110 \end{array}$$

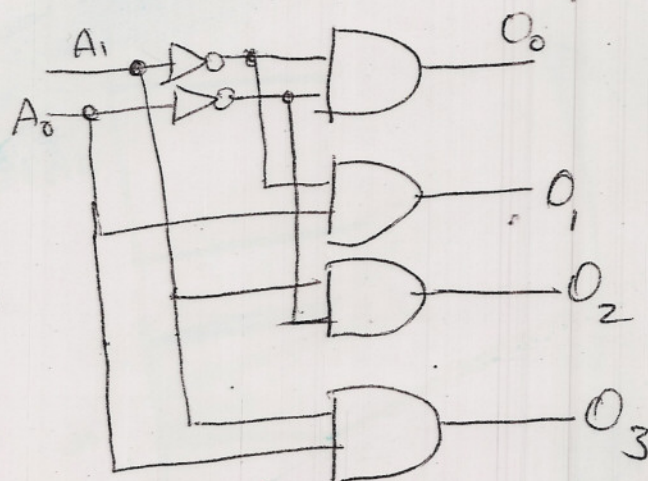
8. Multiply unsigned binary numbers 1010 with 10. (1 Mark)

$$\begin{array}{r} 1010 \times 10 \\ \hline 10100 \\ \hline 101000 \\ \hline 1010000 \end{array} \quad \text{Decimal } 10 \times 2 = 20$$

9. Design the logic circuit for a 1 of 4 decoder. The outputs are active High and the inputs are also active High. (3 Marks)

Function Table

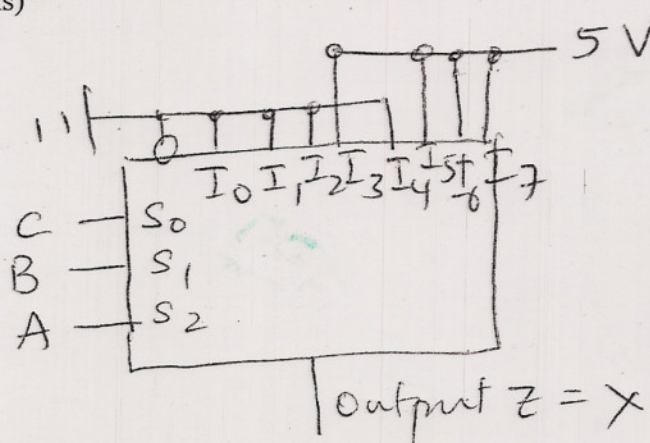
A_1	A_0	Output
0	0	$O_0 = H$, Rest Low
0	1	$O_1 = H$, Rest Low
1	0	$O_2 = H$, Rest Low
1	1	$O_3 = H$, Rest Low



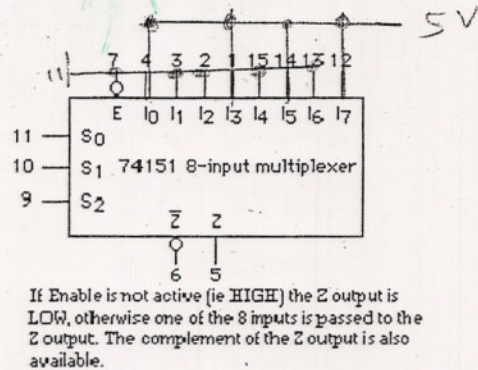
10. Using a 8 input Multiplexer 74HC151 to realize the logic function

$X = AB + BC + AC$ (3 Marks)

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1 ✓
1	0	0	0
1	0	1	1 ✓
1	1	0	1 ✓
1	1	1	1 ✓

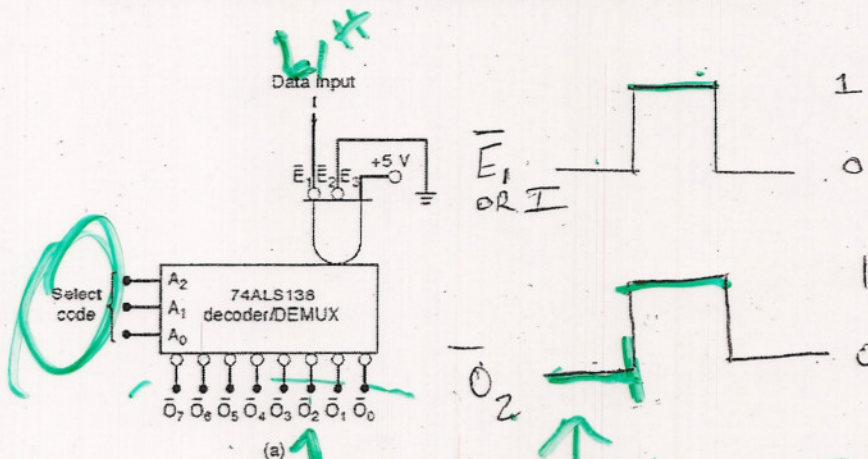


11. It is given that the inputs $I_0 I_3 I_5 I_7$ of 74HC151 are connected to 5V and the rest of the inputs are connected to ground. The E line is connected to ground. Write the function table of the Multiplexer considering Z as the output (ignore inverse of Z output). (3 Marks)



S_2	S_1	S_0	Z
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

12. It is given that the select code applied to the 74LS138 decoder is $A_2 A_1 A_0 = 010$, Determine the waveforms at the outputs. (3 Marks)

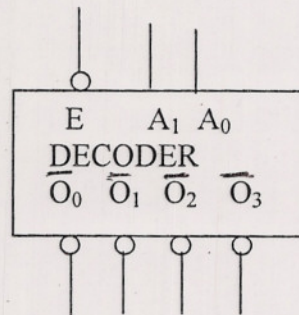


↓
Means output $\bar{O}_2$ is selected

↑ = Rest of the outputs $\bar{O}_0, \bar{O}_1, \bar{O}_3 \rightarrow \bar{O}_7$ are High

13. Assume you are given 5 numbers of 1 of 4 decoder that has active low outputs and active high inputs and active low enable as shown below. Design a 1 of 16 decoder with the given decoders. (3 Marks)

4 marks
16 outputs



DCBA	Outputs
0000	$\bar{O}_0$ alone Low
0001	$\bar{O}_1$ alone Low
0010	$\bar{O}_2$ alone Low
0011	$\bar{O}_3$ alone Low
0100	$\bar{O}_4$ alone Low
0101	$\bar{O}_5$
0110	$\bar{O}_6$
0111	$\bar{O}_7$
1000	$\bar{O}_8$
1001	$\bar{O}_9$
1010	
1011	
1100	
1101	
1110	
1111	

FUNCTION TABLE

INPUTS			OUTPUTS			
E	A0	A1	$\bar{O}_0$	$\bar{O}_1$	$\bar{O}_2$	$\bar{O}_3$
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	H	L	H	L	H	H
L	L	H	H	H	L	H
L	H	H	H	H	H	L

NOTES:

H = High voltage level
L = Low voltage level
X = Don't care

1 of 16 Decoder

